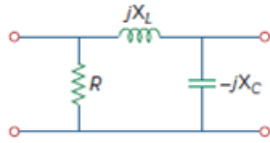


Problem

Using Fig. 19.68, design a problem to help other students better understand how to determine z parameters from an electrical circuit.

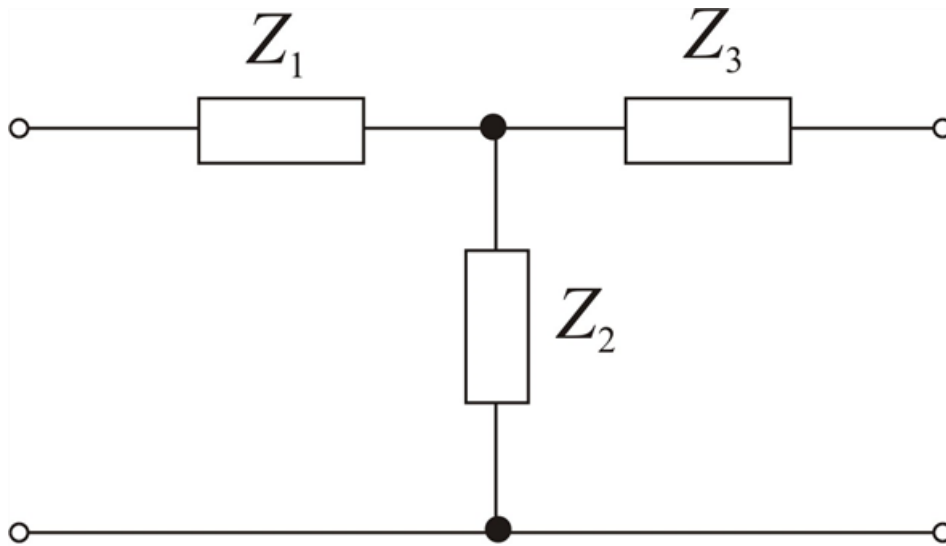
Figure 19.68



Step-by-step solution

Step 1 of 4

Given π network can be transformed to T network.



Step 2 of 4

$$Z_1 = \frac{(R)(jX_L)}{R + jX_L - jX_C}$$

$$Z_1 = \frac{jRX_L}{R + j(X_L - X_C)}$$

$$Z_2 = \frac{(R)(-jX_C)}{R + j(X_L - X_C)}$$

$$Z_2 = \frac{-jRX_C}{R + j(X_L - X_C)}$$

$$Z_3 = \frac{(jX_L)(-jX_C)}{R + j(X_L - X_C)}$$

$$Z_3 = \frac{X_L X_C}{R + j(X_L - X_C)}$$

Step 3 of 4

The impedance parameters are

$$V_1 = Z_{11}I_1 + Z_{22}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

By applying KVL to first loop

$$V_1 = I_1(Z_1 + Z_2) + I_2Z_2$$

By applying KVL to second loop

$$V_2 = I_2(Z_3 + Z_2) + I_1Z_2$$

By comparing these equations with generalized Z- parameters equation

$$Z_{11} = Z_1 + Z_2$$

$$Z_{12} = Z_{21} = Z_2$$

$$Z_{22} = Z_1 + Z_3$$

Step 4 of 4

$$\therefore Z_{12} = Z_{21} = \frac{-jRX_c}{R + j(X_L - X_C)}$$

Therefore

$$Z_{11} = \frac{jRX_L}{R + j(X_L - X_C)} + \frac{-jRX_c}{R + j(X_L - X_C)}$$

$$\therefore Z_{11} = \frac{jR(X_L - X_C)}{R + j(X_L - X_C)}$$

$$Z_{22} = \frac{X_L X_C}{R + j(X_L - X_C)} + \frac{-jRX_c}{R + j(X_L - X_C)}$$

$$\therefore Z_{22} = \frac{X_C(X_L - jR)}{R + j(X_L - X_C)}$$

Thus the matrix form is

$$[Z] = \begin{bmatrix} \frac{jR(X_L - X_C)}{R + j(X_L - X_C)} & \frac{-jRX_c}{R + j(X_L - X_C)} \\ \frac{-jRX_c}{R + j(X_L - X_C)} & \frac{X_C(X_L - jR)}{R + j(X_L - X_C)} \end{bmatrix}$$